GEETHANJALI INSTITUTE OF SCIENCE & TECHNOLOGY
(ANAUTONOMOUSINSTITUTION)
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QUESTIONBANK(DESCRIPTIVE)

Subject Name: Big Data Analytics
SubjectCode:23A3214T
Course &Branch: B.Tech C S (D S)
Year & Semester: IV B.Tech I Semester **Regulation:**RG23

Unit1

S. No	2MarksQuestions(Short)	[BT Level][CO][Marks]
1.	Define Big Data?	L1,CO1,2M
2.	Define structured, semi-structured, and unstructured data?	L1,CO1,2M
3.	Define Web Analytics and mention any two industry applications of Big Data?	L1,CO1,2M
4.	What is Big Data in Marketing? Mention its role in Advertising?	L1,CO1,2M
5.	Define Fraud Analytics, and Credit Risk Management?	L1,CO1,2M
6.	What is Algorithmic Trading?	L1,CO1,2M
7	Name any four Big Data technologies?	L1,CO2,2M
8.	Define Hadoop. Mention any two advantages of Hadoop.	L1,CO2,2M
9	What are Open Source Technologies?	L1,CO2,2M
10	Define Cloud Computing &Firewall analytics?	L1,CO2,2M

S. No	Descriptive Questions(Long)	[BT Level] [CO][Marks]
1	Explain the concept of Big Data and discuss the need for Big Data in modern organizations.	L2,CO1,10M
2	Describe the convergence of key trends that led to the emergence of Big Data. Explain its (5Vs) of Big Data with suitable examples.	L2,CO1,10M
3	Explain various industry examples and applications of Big Data. Discuss the role of Big Data in different business sectors.	L2,CO1,10M
4	Explain the concept of Web Analytics. Discuss how Big Data is used in Marketing and Advertising with suitable examples.	L2,CO1,10M
5	Explain the concept of Fraud Analytics. Discuss the role of Big Data in Fraud Detection with suitable examples.	L2,CO2,10M
6	Discuss the applications of Big Data in Risk Management and Credit Risk Management. Explain their significance in financial institutions.	L2,CO2,10M
7	Explain the concept of Algorithmic Trading. Discuss the role of Big Data in automated trading systems with suitable examples.	L2,CO2,10M

8	Discuss the applications of Big Data in Healthcare and Medicine. Explain how Big Data improves patient care, diagnosis, and medical research.	L2,CO2,10M
9	Describe various Big Data Technologies. Explain the Hadoop framework and its Architecture with Suitable Examples?	L2,CO2,10M
10	Write a Short Note on Mobile Business Intelligence and Crowd Sourcing Analytics in BDA with suitable examples.	L2,CO2,10M
11	Discuss the importance of Open Source Technologies. Explain the relationship between Cloud Computing and Big Data.	L2,CO2,10M
12	Write a Short Note on Inter-firewall Analytics and Trans-firewall Analytics in BDA with Suitable Examples?	L2,CO2,10M

Unit-2

S. No	2MarksQuestions(Short)	[BT Level][CO][Marks]
1.	Define NoSQL	L1,CO3,2M
2.	What are aggregate data models?	L1,CO3,2M
3.	Define Key-Value data model and Document data model.	L1,CO3,2M
4.	What are relationships in NoSQL databases?	L1,CO3,2M
5.	Define Schema-less Database and Materialized View.	L1,CO3,2M
6.	What is a distribution model?	L1,CO3,2M
7.	Differentiate between Master-Slave Replication and Peer-to-Peer Replication.	L1,CO3,2M
8.	What is sharding and replication?	L1,CO3,2M
9	What is meant by relaxing consistency and version stamps?	L1,CO3,2M
10	Write the steps involved in creating a table, loading data, and reading data in Cassandra.	L1,CO3,2M

S.No	Descriptive Questions(Long)	[BT Level] [CO][Marks]
1	Explain the concept of NoSQL databases. Discuss the advantages and limitations of NoSQL databases over traditional relational databases.	L2,CO3,10M
2	Explain the Aggregate Data Model?&Discuss the concept of its Aggregates with suitable examples.	L2,CO3,10M
3	Write a short Note on Key-Value Data Model and Document Data Model with Suitable Examples?Explain it?	L2,CO3,10M
4	Explain the relationships in NoSQL databases& Graph Databases with a neat sketch?	L3,CO3,10M
5	Write a short note on Schema-less Databases & Materialized Views in NoSQL databases with suitable Examples?	L2,CO3,10M
6	Discuss the Distribution Models used in NoSQL databases. Explain the concept of Sharding and its significance in distributed database systems.	L2,CO3,10M
7	Explain the Master-Slave Replication model. Compare it with the Peer-to-Peer Replication model with suitable diagrams.	L2,CO3,10M
8	Discuss the concepts of Sharding and Replication. Explain how they improve scalability, availability, and fault tolerance in distributed databases.	L2,CO3,10M
9	Explain the concept of Consistency in distributed databases. Discuss Relaxing Consistency and Version Stamps with suitable examples.	L2,CO3,10M
10	Explain the architecture of Apache Cassandra. Discuss its role as a distributed NoSQL database.	L2,CO3,10M
11	Explain the process of creating tables, loading data, and reading data	L2,CO3,10M

	in Cassandra with suitable CQL commands and its examples.	
12	Write a short Note on Apache Cassandra by explaining data storage, replication, consistency, query processing, and practical applications.	L2,CO3,10M

Unit-3

S. No	2MarksQuestions(Short)	[BT Level][CO][Marks]
1.	Define data formats.	L1,CO4,2M
2.	What is Hadoop?	L1,CO4,2M
3.	Define HDFS	L1,CO4,2M
4.	What is fault tolerance?	L1,CO4,2M
5.	Define High Availability and Data Locality in Hadoop.	L1,CO4,2M
6.	What is MapReduce?	L1,CO4,2M
7.	What is Apache Hive?	L1,CO4,2M
8.	Define DDL&DML?	L1,CO4,2M
9	Define HiveQL	L1,CO4,2M
10	Define Logical Joins,	L1,CO4,2M

S. No	Descriptive Questions(Long)	[BT Level][CO][Marks]
1	Explain Hadoop data formats, data analysis, and scaling out with a neat Diagram?	L2,CO4,10M
2	Explain HDFS architecture, fault tolerance, data replication, high availability, and data locality in Hadoop?	L2,CO4,10M
3	Explain MapReduce architecture and its process flow with a neat diagram.	L2,CO4,10M
4	Explain the Java interface and data flow of MapReduce with a suitable example.	L2,CO4,10M
5	Explain Hadoop I/O and data integrity, including checksums with a neat Diagram?	L2,CO4,10M
6	Discuss data compression and serialization techniques in Hadoop Ecosystem	L2,CO4,10M
7	Explain Hive architecture, data types, and its file formats with a suitable Example?	L2,CO4,10M
8	Write a short Note on HiveQL data definition (DDL) and data manipulation (DML) with suitable examples.	L2,CO4,10M
9	Explain logical joins and window functions in HiveQL with Suitable examples.	L4,CO4,10M
10	Explain Hive query optimization techniques, including table partitioning and bucketing In Hadoop?	L2,CO4,10M
11	Write a short Note on Indexing in Hive and Hive Join Strategies and their Impact on query Performance?	L2,CO4,10M

Unit-4

S. No	2MarksQuestions(Short)	[BT Level][CO][Marks]
1.	Define Apache Spark.	L1,CO5,,2M
2.	What are Lazy Evaluation and In-Memory Processing in Apache Spark?	L1,CO5,2M
3.	Define Directed Acyclic Graph (DAG)	L1,CO5,2M
4.	What is a Resilient Distributed Dataset (RDD)?	L1,CO5,2M
5.	Define Actions and DataFrames.	L1,CO5,2M
6.	Define Actions and DataFrames.	L1,CO5,2M
7.	What are Complex Data Types and JSON in Apache Spark?	L1,CO5,2M
8.	Define Grouping, Window Functions, and Joins in Spark SQL.	L1,CO5,2M
9	What are Data Sources, Broadcast Variables, and Accumulators in Apache Spark?	L1,CO5,2M
10	What are the different deployment modes of Apache Spark?	L1,CO5,2M

S.No	Descriptive Questions(Long)	[BT Level][CO][Marks]
1	Explain Apache Spark and its advantages over Hadoop. Discuss lazy evaluation, in-memory processing, and DAG execution in Real time?	L2,CO5,10M
2	Explain SparkContext and SparkSession, including their roles in Spark applications?	L2,CO5,10M
3	Explain RDDs transformations-narrow and wide transformations, and actions with suitable examples.	L2,CO5,10M
4	Explain DataFrames and the conversion of RDDs to DataFrames. Discuss the Catalyst Optimizer and its role in query optimization.	L2,CO5,10M
5	Explain DataFrame transformations and operations for handling dates, timestamps, and null values in Apache Spark.	L2,CO5,10M
6	Explain how Spark handles complex data types and JSON data with suitable examples.	L2,CO5,10M
7	Write a short Note on grouping, window functions, and joins in Spark DataFrames with suitable examples.	L4,CO5,10M
8	Discuss Spark data sources and explain how to read and write data using different formats.	L2,CO5,10M
9	Write a short Note On variables and accumulators in Apache Spark with suitable examples.	L2,CO5,10M

10	Explain Spark deployment on on-premises clusters. Discuss cluster managers, Standalone Mode, and Spark on YARNin Spark?	L2,CO5,10M
11	Write a short Note on Spark logs, the Spark UI, and the Spark History Server for monitoring Spark applications.	L2,CO5,10M
12	Discuss debugging techniques and Spark First Aid for identifying and resolving common Spark application problems.	L2,CO5,10M

Unit-5

S.No	2MarksQuestions(Short)	[BTLevel][CO][Marks]
1	Define Spark Performance Tuning	L1,CO6,2M
2	What is Stream Processing?	L1,CO6,2M
3	Define Event Time and Stateful Processing in Structured Streaming.	L1,CO6,2M
4	What is a Window operation?	L1,CO6,2M
5	What are Watermarks?	L1,CO6,2M
6	What is meant by dropping duplicates in a stream?	L1,CO6,2M
7	Define Structured Streaming. List its core concepts.	L1,CO6,2M
8	What are Transformations on Streams?	L1,CO6,2M
9	What are Input Sources and Output Sinks in Structured Streaming?	L1,CO6,2M
10	Mention any four real-world applications of Spark Structured Streaming.	L1,CO6,2M

S. No	Descriptive Questions(Long)	[BT Level][CO][Marks]
1	Explain Spark performance tuning techniques and discuss methods to improve the performance of Spark applications.	L4,C06,10M
2	Explain the fundamentals of stream processing and compare stream processing with batch processing.	L4,C06,10M
3	Write a short Note on event-time processing and stateful processing in Apache Spark Structured Streaming with suitable examples.	L2,C06,10M
4	Explain windows on event time and discuss tumbling windows with suitable examples.	L2,C06,10M
5	Explain watermarks in Structured Streaming and discuss how late-arriving data is handled.	L2,C06,10M
6	Explain the techniques for dropping duplicate records in a stream using Structured Streaming.	L2,C06,10M
7	Explain the core concepts and architecture of Spark Structured Streaming with a suitable diagram.	L4,C06,10M
8	Explain Structured Streaming in action, illustrating how to build and execute a streaming application.	L2,C06,10M
9	Explain different transformations on streams in Structured Streaming with suitable examples.	L4,CO6,10M
10	Explain the input and output operations in Structured Streaming, including supported sources, sinks, and output modes.	L2,CO6,10M

Signature of the Staff:

Signature of Department Academic Committee Member 1:

Signature of Department Academic Committee Member 2:

Signature of Department Academic Committee Member 3: